2 Gaussian-chaos decomposition

We use the probabilists' Hermite polynomials

$$H_\alpha(G) = \prod_{I\in\mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I), \qquad \mathbb{E}[H_\alpha H_\eta] = \alpha!\,\mathbf{1}_{\{\alpha=\eta\}},$$

where $\alpha! = \prod_I \alpha_I!$. Since the H_α form a complete orthogonal basis of $L^2(\gamma_{|\mathcal{I}_N|})$, where $\gamma_{|\mathcal{I}_N|}$ denotes standard Gaussian measure on $\mathbb{R}^{|\mathcal{I}_N|}$, we have

$$Z_N(\beta : G) = \sum_{k=0}^{\infty} \sum_{\alpha:|\alpha|=k} \frac{\mathbb{E}[Z_N(\beta : G')H_\alpha(G')]}{\alpha!} H_\alpha(G), \tag{1}$$

where the second sum runs over all multi-index α such that $\sum_I \alpha_I = k$, and the expectation is taken over G' independent copy of G. Let P_r denote the orthogonal projection in $L^2(\gamma_{|\mathcal{I}_N|})$ onto the r-th homogeneous Gaussian chaos: Then, we can write

$$P_r Z_N(\beta; G) = \sum_{|\alpha|=r} \frac{\mathbb{E}[Z_N(\beta; G')H_\alpha(G')]}{\alpha!} H_\alpha(G).$$

Since $\mathbb{E}Z_N(\beta)$ is deterministic, we also have

$$P_r Y_N(\beta) = \frac{P_r Z_N(\beta)}{\mathbb{E}Z_N(\beta)}.$$

By the same orthogonality and Lemma 2.2, for every sufficiently regular random variable $X = X(G)$,

$$\|P_r X\|_2^2 = \sum_{|\alpha|=r} \frac{|\mathbb{E}\,\partial_g^\alpha X|^2}{\alpha!}.$$

Recall $A_N(\beta) := \frac{\beta}{2}\,\partial_\beta \log \mathbb{E}Z_N(\beta)$ from (**??**). The next lemma proves the exact radial decomposition

$$Y_N(\beta) - 1 = A_N(\beta)Q_N + \mathcal{R}_N(\beta), \qquad \mathcal{R}_N = \sum_{r\geq 3} P_r Y_N.$$

2.1 Exact finite-size radial projection

Call a multi-index α *even* if

$$\sum_{I\ni x} \alpha_I \equiv 0 \pmod 2 \qquad \text{for every } x \in [N]. \tag{2}$$

Otherwise, we call α odd.

Lemma 2.1. *For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$, define $\chi_s(I) = \prod_{x\in I} s_x$. We have*

$$Z_N(\beta; G) = Z_N\big(\beta; (\chi_s(I)g_I)_I\big). \tag{3}$$

In particular, if α is odd, then

$$\mathbb{E}[Z_N(\beta; G)H_\alpha(G)] = 0.$$

Proof. To make the Clifford symmetry explicit, let

$$\Gamma := \psi_1 \cdots \psi_N.$$

Note that $\Gamma^*\Gamma$ is identity map, so Γ is unitary. Since N is even, moving ψ_j through the other $N-1$ Majorana generators produces the sign $(-1)^{N-1} = -1$. Hence

$$\Gamma\psi_j\Gamma^* = -\psi_j \qquad (j \in [N]).$$

For each $x \in [N]$, put $U_x = \Gamma\psi_x$. This is unitary, and conjugation by U_x is the composition of conjugation by ψ_x and conjugation by Γ. The Clifford relations give

$$\psi_x\psi_j\psi_x = \begin{cases} \psi_x, & j = x, \\ -\psi_j, & j \neq x. \end{cases}$$

Combining the last two displays, we obtain

$$U_x\psi_j U_x^* = \begin{cases} -\psi_x, & j = x, \\ \psi_j, & j \neq x. \end{cases}$$

Thus conjugation by U_x changes the sign of precisely the generator ψ_x.

Let $S_s = \{x \in [N] : s_x = -1\}$ and let U_s be the product of the unitaries U_x over $x \in S_s$, in any fixed order. Successively applying the preceding conjugation rule yields

$$U_s\psi_j U_s^* = s_j\psi_j \qquad (j \in [N]).$$

If $I = \{i_1 < \cdots < i_q\}$, then conjugation is multiplicative, so the definition $\psi_I = \mathrm{i}^{q/2}\psi_{i_1}\cdots\psi_{i_q}$ gives

$$\begin{aligned} U_s\psi_I U_s^* &= \mathrm{i}^{q/2}(U_s\psi_{i_1}U_s^*)\cdots(U_s\psi_{i_q}U_s^*) \\ &= \left(\prod_{x\in I} s_x\right)\psi_I = \chi_s(I)\psi_I. \end{aligned}$$

Writing $g^{(s)} = (\chi_s(I)g_I)_I$, we therefore have

$$\begin{aligned} H_N(g^{(s)}) &= \sigma_N \sum_{I\in\mathcal{I}_N} \chi_s(I) g_I \psi_I \\ &= \sigma_N \sum_{I\in\mathcal{I}_N} g_I U_s\psi_I U_s^* = U_s H_N(G) U_s^*. \end{aligned}$$

The power-series definition of the exponential then implies

$$e^{-\beta H_N(g^{(s)})} = U_s e^{-\beta H_N(G)} U_s^*.$$

Finally, the normalized trace is invariant under unitary conjugation. Consequently

$$Z_N(\beta; g^{(s)}) = \operatorname{tr}\left(U_s e^{-\beta H_N(G)} U_s^*\right) = \operatorname{tr} e^{-\beta H_N(G)} = Z_N(\beta; G),$$

which proves (3).

Assume that α is odd, i.e. we can take x such that $\sum_{I:x\in I}\alpha_I$ is odd. We take s such that $s_y = 1$ for $y \neq x$ and $s_x = -1$. Then,

$$H_\alpha((\chi_s(I)g_I)_I) = \left(\prod_{x=1}^{N} s_x^{\sum_{I\ni x}\alpha_I}\right) H_\alpha(G) = -H_\alpha(G).$$

Hence, we have

$$\mathbb{E}[Z_N(G')H_\alpha(G')] = \mathbb{E}[Z_N(\chi_s G')H_\alpha(\chi_s G')] = -\mathbb{E}[Z_N(G')H_\alpha(G')],$$

which gives the second claim. □

2.2 Gaussian integration by parts

Lemma 2.2. *Let $f : \mathbb{R}^{|\mathcal{I}_N|} \to \mathbb{R}$ be of class $C^{|\alpha|}$, and suppose that f and its mixed partial derivatives up to order α have at most exponential growth in $\|G\|$. Then*

$$\mathbb{E}[H_\alpha(G)f(G)] = \mathbb{E}\big[\partial_g^\alpha f(G)\big]\,. \tag{4}$$

In particular, (4) holds for $f(G) = Z_N(\beta; G)$.

Proof. Recall that the one-dimensional integration-by-parts identity yields

$$\mathbb{E}[\mathrm{He}_k(g)f(g)] = \mathbb{E}\Big[f^{(k)}(g)\Big]\,, \qquad g \sim N(0,1),$$

for every function $f \in C^k(\mathbb{R})$ satisfying the stated growth condition. Applying this identity successively to the independent Gaussian coordinates $(g_I)_{I\in\mathcal{I}_N}$ and using

$$H_\alpha(G) = \prod_{I\in\mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I),$$

gives

$$\begin{aligned}
\mathbb{E}[H_\alpha(G)f(G)] &= \mathbb{E}\left[\left(\prod_{I\in\mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I)\right) f(G)\right] \\
&= \mathbb{E}\left[\left(\prod_{I\in\mathcal{I}_N} \partial_{g_I}^{\alpha_I}\right) f(G)\right] \\
&= \mathbb{E}\big[\partial_g^\alpha f(G)\big]\,.
\end{aligned}$$

The growth assumption justifies all integrations by parts. Finally, for fixed N and β, the partition function $Z_N(\beta; G)$ is a smooth function of finitely many Gaussian coordinates, and all of its derivatives have at most exponential growth in $\|G\|$. Thus it satisfies the assumption. □

2.3 First and second moment

Lemma 2.3. *For every real β,*

$$P_1 Y_N(\beta) = 0$$

and

$$P_2 Y_N(\beta) = \frac{\beta\,\partial_\beta \log \mathbb{E}Z_N(\beta)}{2M_N} \sum_{I\in\mathcal{I}_N} (g_I^2 - 1). \tag{5}$$

Proof. This parity rule above kills the first chaos. In the second chaos it also kills every off-diagonal monomial $g_I g_J$, since the parity condition for such a monomial forces $I = J$. Permutation symmetry shows that the remaining diagonal coefficients agree. Consequently

$$P_2 Z_N(\beta) = b_N(\beta) \sum_I (g_I^2 - 1)$$

for a deterministic scalar b_N.

It remains to compute the coefficient. For each fixed I, the probabilists' second Hermite polynomial is $g_I^2 - 1$, with

$$\mathbb{E}(g_I^2 - 1)^2 = 2.$$

Since P_2 is an orthogonal projection and the variables g_I are independent, taking the inner product of the preceding formula for $P_2 Z_N$ with $g_I^2 - 1$ gives

$$2b_N(\beta) = \mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big].$$

Applying Gaussian integration by parts twice, conditionally on all coordinates other than g_I, yields

$$\mathbb{E}\big[Z_N(\beta)(g_I^2-1)\big] = \mathbb{E}\, g_I \partial_{g_I} Z_N(\beta).$$

Summing over the $\mathcal{I}_N$ choices of I therefore gives the first equality below. A further conditional Gaussian integration by parts gives the second:

$$\begin{aligned} 2M_N b_N(\beta) &= \sum_I \mathbb{E}\big[g_I \partial_{g_I} Z_N(\beta)\big] \\ &= \beta\, \partial_\beta \mathbb{E} Z_N(\beta) = \beta F_N'(\beta). \end{aligned}$$

The last equality follows pointwise from the exact scaling identity

$$Z_N(\beta; tG) = \operatorname{tr} e^{-\beta H_N(tG)} = \operatorname{tr} e^{-t\beta H_N(G)} = Z_N(t\beta; G),$$

where we used $H_N(tG) = tH_N(G)$. Differentiating at $t=1$ gives

$$\sum_I g_I \partial_{g_I} Z_N(\beta; G) = \beta\, \partial_\beta Z_N(\beta; G).$$

This proves (5). □

2.4 Direct Rademacher control of all higher chaoses

Lemma 2.4. *For real β, every multi-index α of total order r satisfies*

$$\big|\partial_g^\alpha Z_N(\beta; G)\big| \le (|\beta|\sigma_N)^r Z_N(\beta; G). \tag{6}$$

Proof. The case $r=0$ is immediate, so assume $r \ge 1$. Write $H_N = H_N(G)$ for brevity. We first derive the equality

$$\begin{aligned} &\partial_{g_{I_r}} \cdots \partial_{g_{I_1}} Z_N(\beta; G) \\ &\quad = (-\beta\sigma_N)^r \sum_{\pi \in S_r} \int_{\Delta_r} \operatorname{tr}\Big(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1 \beta H_N} \cdots \psi_{I_{\pi(r)}} e^{-t_r \beta H_N} \Big)\, \mathrm{d}t_1 \cdots \mathrm{d}t_r \end{aligned} \tag{7}$$

where

$$\Delta_r = \left\{ (t_j)_{j=1}^r : t_j \ge 0, \quad \sum_{j=1}^r t_j \le 1 \right\}, \qquad t_0 = 1 - \sum_{i=1}^r t_j.$$

For a differentiable matrix-valued function $A(\theta)$, the one-variable Duhamel formula is

$$\frac{\mathrm{d}}{\mathrm{d}\theta} e^{A(\theta)} = \int_0^1 e^{sA(\theta)} A'(\theta) e^{(1-s)A(\theta)}\, \mathrm{d}s. \tag{8}$$

By the definition of H_N in Subsection **??**,

$$\partial_{g_I} H_N = \sigma_N \psi_I.$$

Applying (8) to $A = -t\beta H_N$ gives, for $t \ge 0$,

$$\begin{aligned} \partial_{g_I} e^{-t\beta H_N} &= -t\beta\sigma_N \int_0^1 e^{-st\beta H_N} \psi_I e^{-(1-s)t\beta H_N}\, \mathrm{d}s \\ &= -\beta\sigma_N \int_0^t e^{-u\beta H_N} \psi_I e^{-(t-u)\beta H_N}\, \mathrm{d}u. \end{aligned} \tag{9}$$

Here, (9) with $t = 1$ yields (7) for $r = 1$. Suppose inductively that the formula holds at order $r - 1$. Then

$$\begin{aligned}&\partial_{g_{I_r}} \cdots \partial_{g_{I_1}} Z_N(\beta; G)\\ &\quad = (-\beta\sigma_N)^{r-1} \sum_{\pi \in S_{r-1}} \int_{\Delta_{r-1}} \partial_{g_{I_r}} \operatorname{tr}\Big(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\pi(r-1)}} e^{-t_{r-1}\beta H_N}\Big)\, \mathrm{d}t_1 \cdots \mathrm{d}t_{r-1}.\end{aligned} \tag{10}$$

For fixed $\pi \in S_{r-1}$ and $j \in \{1, \dots, r\}$, differentiating the factor $e^{-t_{j-1}\beta H_N}$ replaces it, by (9), with

$$-\beta\sigma_N \int_0^{t_{j-1}} e^{-u\beta H_N} \psi_{I_r} e^{-(t_{j-1}-u)\beta H_N}\, \mathrm{d}u.$$

Splitting t_{j-1} into u and $t_{j-1} - u$ and relabeling the resulting simplex variables, the corresponding term in the Leibniz rule becomes

$$-\beta\sigma_N \int_{\Delta_r} \operatorname{tr}\Big(e^{-t_0\beta H_N} \psi_{I_{\sigma_{\pi,j}(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\sigma_{\pi,j}(r)}} e^{-t_r\beta H_N}\Big)\, \mathrm{d}t_1 \cdots \mathrm{d}t_r,$$

where

$$\sigma_{\pi,j}(k) = \begin{cases} \pi(k) & (k < j) \\ r & (k = j) \\ \pi(k-1) & (k > j) \end{cases}.$$

Since $(\pi, j) \mapsto \sigma_{\pi,j}$ is a bijection from $S_{r-1} \times \{1, \dots, r\}$ onto S_r, summing the Leibniz-rule terms in (10) proves (7).

It remains to estimate each integrand in (7). Here and below the Schatten norms are defined using the normalized trace:

$$\|A\|_p = (\operatorname{tr}|A|^p)^{1/p}, \qquad \|A\|_\infty = \|A\|.$$

For $t_j > 0$,

$$\left\| e^{-t_j\beta H_N} \right\|_{1/t_j} = \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j},$$

with the usual $p = \infty$ interpretation when $t_j = 0$. By Subsection **??**, $\|\psi_I\|_\infty = 1$. Noncommutative Hölder, with Schatten exponents $1/t_j$, therefore gives, for every $\pi \in S_r$ and $t \in \Delta_r$,

$$\begin{aligned}&\left|\operatorname{tr}\Big(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\pi(r)}} e^{-t_r\beta H_N}\Big)\right| \\ &\quad \le \prod_{j=0}^{r} \left\| e^{-t_j\beta H_N} \right\|_{1/t_j} \prod_{k=1}^{r} \left\| \psi_{I_{\pi(k)}} \right\|_\infty \\ &\quad \le \prod_{j=0}^{r} \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j} = \operatorname{tr} e^{-\beta H_N} = Z_N(\beta; G),\end{aligned} \tag{11}$$

where we used $\sum_{j=0}^{r} t_j = 1$. Combining (7) and (11), we obtain

$$\begin{aligned}\left|\partial_{g_{I_r}} \cdots \partial_{g_{I_1}} Z_N(\beta; G)\right| &\le (|\beta|\sigma_N)^r \sum_{\pi \in S_r} \int_{\Delta_r} Z_N(\beta; G)\, \mathrm{d}t_1 \cdots \mathrm{d}t_r \\ &= (|\beta|\sigma_N)^r Z_N(\beta; G) |S_r| \int_{\Delta_r} \mathrm{d}t_1 \cdots \mathrm{d}t_r \\ &= (|\beta|\sigma_N)^r Z_N(\beta; G),\end{aligned}$$

which proves (6). □

Recall that Lemma 2.1 shows that only even multi-indices have nonzero Hermite coefficients.

Let $(\varepsilon_x)_{x=1}^N$ be independent Rademacher variables on an auxiliary probability space, and write $\mathbb{E}_\varepsilon$ for expectation on this space. Set

$$S_{N,q} = \sum_{I \in \mathcal{I}_N} \varepsilon_I, \qquad \varepsilon_I = \prod_{x \in I} \varepsilon_x. \tag{12}$$

Thus $S_{N,q}$ is the degree-q elementary symmetric polynomial in the Rademacher signs.

Lemma 2.5. *For every integer $r \geq 0$,*

$$\sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!} = \frac{1}{r!} \mathbb{E}_\varepsilon \left[S_{N,q}^r \right]. \tag{13}$$

Moreover, there are constants $c_q > 0$ and $C_q < \infty$, depending only on q, such that

$$\mathbb{E}_\varepsilon \left[|S_{N,q}|^3 e^{t|S_{N,q}|} \right] \leq C_q N^{3q/2} \tag{14}$$

for every N and every $0 \leq t \leq c_q N^{-(q-1)}$.

Proof. Starting from the definition (12) and applying the multinomial formula gives

$$\frac{1}{r!} \mathbb{E}_\varepsilon \left[S_{N,q}^r \right] = \frac{1}{r!} \mathbb{E}_\varepsilon \left[\sum_{|\alpha|=r} r! \prod_{I \in \mathcal{I}_N} \frac{\varepsilon_I^{\alpha_I}}{\alpha_I!} \right] = \sum_{|\alpha|=r} \frac{1}{\alpha!} \mathbb{E}_\varepsilon \left[\prod_{I \in \mathcal{I}_N} \varepsilon_I^{\alpha_I} \right]. \tag{15}$$

For a fixed α, the product inside the expectation is

$$\prod_{x=1}^N \varepsilon_x^{\sum_{I : x \in I} \alpha_I}.$$

By independence of the Rademacher variables and the definition of an even multi-index in (2), its expectation is one if α is even and zero otherwise. Substitution into (15) proves (13).

We next prove a deterministic bound for $S_{N,q}$. For every $N \geq 1$ and every integer $m \geq 0$, let $e_m^{(N)}$ denote the mth elementary symmetric polynomial in N variables, that is,

$$e_m^{(N)}(z_1, \ldots, z_N) := \sum_{\substack{I \subset [N] \\ |I|=m}} \prod_{x \in I} z_x, \qquad e_0^{(N)}(z_1, \ldots, z_N) = 1.$$

For the rest of the proof, we use the abbreviation

$$e_m := e_m^{(N)}(\varepsilon_1, \ldots, \varepsilon_N).$$

In particular, $e_1 = \sum_{x=1}^N \varepsilon_x$, and (12) gives $S_{N,q} = e_q$. Newton's identities state that

$$m e_m = \sum_{j=1}^m (-1)^{j-1} e_{m-j} p_j, \tag{16}$$

where $p_j := \sum_{x=1}^N \varepsilon_x^j$. Since $\varepsilon_x \in \{\pm 1\}$, we have

$$p_j = \begin{cases} e_1, & j \text{ odd}, \\ N, & j \text{ even}. \end{cases}$$

An induction on m, using (16), shows that there is a constant $C_m < \infty$ such that

$$|e_m| \le C_m \left(N^{m/2} + |e_1|^m \right).$$

Indeed, suppose that the estimate is known below order m and set

$$X_N := \frac{e_1}{\sqrt{N}}.$$

For every $j \ge 1$,

$$|p_j| \le N^{j/2}(1 + |X_N|^j).$$

Hence, with $C_0 = 1$,

$$\begin{aligned} |e_{m-j} p_j| &\le C_{m-j} N^{m/2} (1 + |X_N|^{m-j})(1 + |X_N|^j) \\ &\le 2C_{m-j} N^{m/2} (1 + |X_N|^m), \end{aligned}$$

where we used $1 + |X|^m - |X|^{m-j} - |X|^j = (1 - |X|^j)(1 - |X|^{m-j}) \ge 0$. Substitution into (16) proves the induction step. Taking $m = q$ and using (12), we obtain

$$|S_{N,q}| \le C_q \left(N^{q/2} + |e_1|^q \right). \tag{17}$$

Since $q \ge 4$ by Subsection **??** and $|X_N| \le \sqrt{N}$, (17) implies that, for $0 \le t \le c_q N^{-(q-1)}$,

$$\begin{aligned} t|S_{N,q}| &\le C_q c_q N^{-(q/2-1)} \left(1 + |X_N|^q\right) \\ &\le C_q c_q \left(1 + |X_N|^2\right). \end{aligned} \tag{18}$$

Indeed,

$$N^{-(q/2-1)} |X_N|^q = |X_N|^2 \left(\frac{|X_N|}{\sqrt{N}} \right)^{q-2} \le |X_N|^2.$$

Choose c_q small enough that the last member of (18) is bounded by $(1 + |X_N|^2)/4$.

Hoeffding's inequality gives

$$\mathbb{P}_\varepsilon(|X_N| \ge u) \le 2e^{-u^2/2} \qquad (u \ge 0). \tag{19}$$

Consequently, integration of the tail in (19) yields

$$\sup_{N \ge 1} \mathbb{E}_\varepsilon \left[\left(1 + |X_N|^{3q}\right) e^{|X_N|^2/4} \right] < \infty. \tag{20}$$

Finally, (17), (18), and (20) give

$$\begin{aligned} \mathbb{E}_\varepsilon \left[|S_{N,q}|^3 e^{t|S_{N,q}|} \right] &\le C_q N^{3q/2} \mathbb{E}_\varepsilon \left[\left(1 + |X_N|^{3q}\right) e^{(1+|X_N|^2)/4} \right] \\ &\le C_q N^{3q/2}, \end{aligned}$$

which proves (14). □

Proposition 2.6. *There exists $\beta_{\mathrm{Rad}} = \beta_{\mathrm{Rad}}(q) > 0$ such that, for every fixed real β with $|\beta| < \beta_{\mathrm{Rad}}$,*

$$\sum_{r \ge 3} \|P_r Y_N(\beta)\|_2^2 = O\left(N^{-3(q/2-1)} \right) = o(N^{2-q}).$$

Proof. Put

$$t_N = \beta^2 \sigma_N^2 = \beta^2 (q-1)! N^{-(q-1)}.$$

By the Hermite expansion (1) and its orthogonality relation, the squared norm of the rth projection is the sum of the squared Hermite coefficients. Lemma 2.1 eliminates every odd multi-index, while Lemma 2.2 replaces each remaining Hermite coefficient by the corresponding expected derivative. Consequently, Lemma 2.4 and $\mathbb{E}Z_N(\beta) > 0$ give

$$\begin{aligned} \|P_r Y_N(\beta)\|_2^2 &= \frac{1}{\mathbb{E}Z_N(\beta)^2} \sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!} \left| \mathbb{E} \partial_g^\alpha Z_N(\beta) \right|^2 \\ &\le t_N^r \sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!}. \end{aligned} \tag{21}$$

By the first assertion of Lemma 2.5, namely (13), summing (21) over all $r \ge 3$ yields

$$\begin{aligned} \sum_{r \ge 3} \|P_r Y_N(\beta)\|_2^2 &\le \sum_{r \ge 3} \frac{t_N^r}{r!} \mathbb{E}_\varepsilon \left[S_{N,q}^r \right] \\ &\le \mathbb{E}_\varepsilon \left[\sum_{r \ge 3} \frac{(t_N |S_{N,q}|)^r}{r!} \right] \\ &\le \frac{t_N^3}{6} \mathbb{E}_\varepsilon \left[|S_{N,q}|^3 e^{t_N |S_{N,q}|} \right]. \end{aligned} \tag{22}$$

Here the second inequality uses $|\mathbb{E}_\varepsilon[S_{N,q}^r]| \le \mathbb{E}_\varepsilon[|S_{N,q}|^r]$, and the last one uses $\sum_{r \ge 3} u^r / r! \le u^3 e^u / 6$ for $u \ge 0$. Choose

$$\beta_{\mathrm{Rad}} = \sqrt{\frac{c_q}{(q-1)!}},$$

where c_q is the constant in Lemma 2.5. If $|\beta| < \beta_{\mathrm{Rad}}$, then $t_N \le c_q N^{-(q-1)}$. Therefore the second assertion of Lemma 2.5, namely (14), together with (22) imply

$$\begin{aligned} \sum_{r \ge 3} \|P_r Y_N(\beta)\|_2^2 &\le C_{q,\beta} N^{-3(q-1)} N^{3q/2} \\ &= C_{q,\beta} N^{-3(q/2-1)}. \end{aligned}$$

Since $q \ge 4$, we have $3(q/2-1) > q-2$, which completes the proof. □